

A Diagnosis Algorithm on the Star Graph under the PMC Model

Cheng-Kuan Lin and Jimmy J. M. Tan
 Department of Computer Science
 National Chiao Tung University
 Hsinchu City, Taiwan 300, R.O.C.
 {cklin, jmtan}@cs.nctu.edu.tw

Yuan-Hsiang Teng*
 Department of Computer Science and Information Engineering
 Hungkuang University
 Taichung City, Taiwan 433, R.O.C.
 yhteng@sunrise.hk.edu.tw

Po-Sheng Pan and Lih-Hsing Hsu
 Department of Computer Science and Information Engineering
 Providence University
 Taichung City, Taiwan 433, R.O.C.
 {g9872009, lhhsu}@pu.edu.tw

Abstract

In this paper, we propose a specific structure in the star graph for local diagnosis under the PMC model. We design an adaptive local diagnosis algorithm for a star graph S_n . We prove that S_n is $(n - 1)$ -diagnosable with our proposed algorithm under the PMC model. Moreover, with our algorithm, a diagnosis is completed in three test rounds.

1 Introduction

A multiprocessor system consists of processors and communication links between processors. In practice, most multiprocessor systems are based on an underlying bus structure, or fabric, and are perfectly feasible for a central test controller (an independent processor acting as a controller) to check each processor in the system. In such a scheme, the central controller itself can be tested externally. Some research is related to the issue of network-on-chip (NoC); for example, Pande et al.

[9] developed an evaluation methodology to compare the performance and characteristics of a variety of NoC topologies; Bartic et al. [2] presented an NoC design which is suitable for building networks with irregular topologies.

The problem of identifying faulty processors in a multiprocessor system has been widely addressed in [3, 4, 5, 6, 8, 10, 11]. Throughout this paper, the underlying topology of a multiprocessor system is modeled as a graph; each processor is represented by a vertex, and the communication bus, or fabric, is represented by a single edge between two vertices. A diagnosis testing signal is supposed to be delivered from one vertex to another one through the communication bus at one time. A system performs a so-called *system-level diagnosis* by making each processor act as a tester to test each of the directly connected ones. It is noticed that such a scheme contains no central test controller instead. Several well-known approaches to system diagnosis have been developed. One classic approach, called the PMC diagnosis model (or PMC model for short), was first proposed by Preparata et al. [10]. This model performs a diagnosis by sending a test signal from a processor to another linked one and then receiving a response

*Corresponding author: yhteng@sunrise.hk.edu.tw

in the reverse direction. According to the collection of all test results, the fault status of every processor can be identified.

In some circumstances, however, we are only concerned about some substructure of a multiprocessor system, which is implementable in very large-scale integration (VLSI). Such a substructure, for example, can be a ring, a path, a tree, a mesh, and so on. If all processors in these substructures can be guaranteed to be fault free, a procedure is still workable even though there are many faulty processors in the remaining part of the system. Thus, the local substructure plays a more critical role than the global status of the entire system. Motivated by such a concept, Hsu and Tan [6] presented an elegant measure of diagnosability, known as local diagnosability, to identify the diagnosability of a system by computing the local diagnosability with respect to each individual processor. For any processor in a system, a useful structure [6] was proposed to determine its local diagnosability under the PMC model.

The n -dimensional star network S_n was proposed in [1] as an attractive alternative to the n -cube topology for interconnecting processors in parallel computers. In this article, we concern with the local diagnosability on the star graph. We give an adaptive local diagnosis algorithm for a star graph with three test rounds. The rest of this article is organized as follows. Section 2 provides preliminary background for system diagnosis and the star graph. Section 3 introduces how to diagnose a star graph with our algorithm. Finally, our conclusions are given in Section 4.

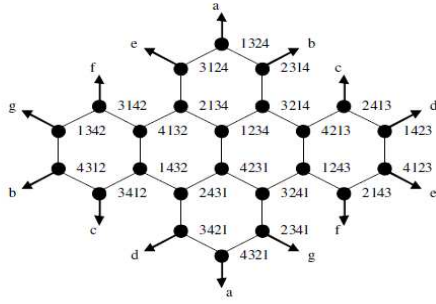
2 Preliminaries

For the graph definitions and notations we follow the standard terminology [7]. Let an undirected graph $G = (V, E)$ denote the underlying topology of a multiprocessor system. Under the PMC diagnosis model [10], we assume that adjacent processors are capable of performing tests on each other. For any two adjacent vertices $u, v \in V$, the ordered pair (u, v) represents a *test* that processor u can diagnose processor v . In this situation, u is a *tester*, and v is a *testee*. The outcome of a test (u, v) is 1 (respectively, 0) if u evaluates v to be faulty (respectively, fault-free). Because the faults considered here are permanent, the outcome of a test is *reliable* if and only if the tester is fault-free. A *test assignment* for system G is a collection of tests and can be modeled as a directed

graph $T = (V, L)$, where $(u, v) \in L$ implies that u and v are adjacent in G . The collection of all test results from the test assignment T is called a *syndrome*. Formally, a syndrome of T is a mapping $\sigma : L \rightarrow \{0, 1\}$. The set F of all faulty processors in G is called a *faulty set*. It is noticed that F can be any subset of V . The process of identifying all faulty vertices is said to be the system diagnosis. Furthermore, the maximum number of faulty vertices that can be correctly identified in a system G is called the *diagnosability* of G , denoted by $\tau(G)$. A system is *t-diagnosable* if, given any syndrome, all faulty units can be identified without replacement, provided that the number of faulty units present does not exceed t [10].

For a multiprocessor system, the random-fault model assumes that the probabilities of processor failures are identical and independent. Let v be any vertex in a graph G . It is intuitive to observe that $(N_G(v), \{v\} \cup N_G(v))$ forms an indistinguishable pair of faulty sets. That is, the conventional diagnosability that has been addressed by many researchers mainly describes the global status of a system under the random-fault model. Instead, Hsu and Tan [6] turned their attention to the local connective substructure in a system. More precisely, given any single vertex v in a graph it is only required to determine whether v is faulty or not. The following concept is proposed in [6]. Let G be a graph and let v be any one of its vertex. Then G is *t*-diagnosable at vertex v* if, given a syndrome σ_F produced by a set of faulty vertex $F \subseteq V$ with $v \in F$ and $|F| \leq t$, every set F' of faulty vertex that is consistent with σ_F must also contain vertex v . The *local diagnosability* of v in G , denoted by $\tau_G^*(v)$, is defined to be the maximum integer of t such that G is *t*-diagnosable at vertex v*.

The star graph is proposed in [1]. We denote an n -dimensional star graph as S_n . The vertex set V of S_n is $\{u_1 u_2 \dots u_n \mid u_1 u_2 \dots u_n \text{ is a permutation of } 1, 2, \dots, n\}$. The adjacency is defined as follows: $u_1 u_2 \dots u_i \dots u_n$ is adjacent to $v_1 v_2 \dots v_i \dots v_n$ through an edge of dimension i with $2 \leq i \leq n$ if $v_j = u_j$ for $j \notin \{1, i\}$, $v_1 = u_i$, and $v_i = u_1$. For example, in a S_4 containing $4!$ vertices, two vertices 1234 and 4231 are neighbors and joined through an edge labeled 4. Figure 1 illustrates the S_4 . It is known that S_n is a bipartite graph with one partite set containing all odd permutations and the other partite set containing all even permutations. For convenience, we refer an even permutation as a white vertex, and refer an odd permutation as a black vertex. By the definition of

Figure 1: A star graph S_4 .

S_n , there is exactly one neighbor $\mathbf{v}$ of $\mathbf{u}$ such that $\mathbf{u}$ and $\mathbf{v}$ are adjacent through an i -dimensional edge with $2 \leq i \leq n$. For this reason, we use $(\mathbf{u})^i$ to denote the unique i -neighbor of $\mathbf{u}$. In this article, we design an adaptive local diagnosis algorithm for a star graph S_n . Moreover, we prove that S_n is $(n-1)$ -diagnosable with our proposed algorithm under the PMC model.

3 A Diagnosis Algorithm on the Star Graph

In this section, we give a three-round adaptive local diagnosis algorithm under the PMC model for the star graph. For any two positive integers r and s , we use $[r]_s$ to denote $r \pmod s$. For simplification, we use $\mathbf{u}_{x,1}, \mathbf{u}_{x,2}, \dots$, and $\mathbf{u}_{x,n-2}$ to denote $(\mathbf{u})^x, ((\mathbf{u})^x)^{[x+1]_n}, \dots$, and $(\dots((\mathbf{u})^x)^{[x+1]_n} \dots)^{[x+n-3]_n}$ for $2 \leq x \leq n$, respectively. To diagnose a vertex in a star graph S_n , we give the following two algorithms, **VOTE** and **SDA**. See Figure 2 for an illustration.

Algorithm VOTE($T, \mathbf{u}$)

Input: A vertex set T and a vertex $\mathbf{u}$.

Output: The value is 0 or 1 if $\mathbf{u}$ is fault-free or faulty, respectively.

Begin

$n_0 \leftarrow 0;$

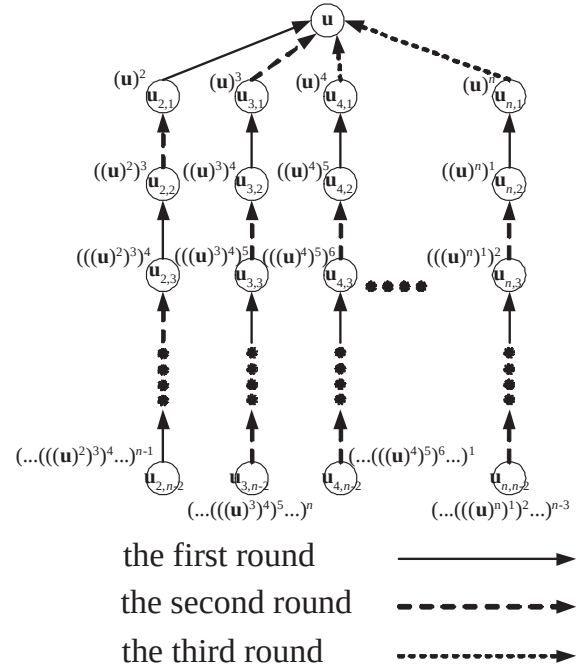
$n_1 \leftarrow 0;$

if $|T| = 1$

then let $\mathbf{x}$ be the vertex in T ;
 return $\sigma(\mathbf{x}, \mathbf{u})$;

else

$n_0 \leftarrow |\{i \mid \sigma(\mathbf{x}_i, \mathbf{u}) = 0 \text{ for each } \mathbf{x}_i \in T\}|$

Figure 2: A diagnosis algorithm on the star graph S_n .

```

        with  $1 \leq i \leq |T|$ };
     $n_1 \leftarrow |\{i \mid \sigma(\mathbf{x}_i, \mathbf{u}) = 1 \text{ for each } \mathbf{x}_i \in T$ 
        with  $1 \leq i \leq |T|$ };
    if  $n_0 \geq n_1$ 
        return 0;
    else
        return 1;
End

```

Algorithm SDA($S_n, \mathbf{u}$)

Input: A star graph S_n and a vertex $\mathbf{u}$.

Output: The value is 0 or 1 if $\mathbf{u}$ is fault-free or faulty, respectively.

Begin

$T \leftarrow \emptyset;$

$p \leftarrow |\{i \mid \sigma(\mathbf{u}_{2,i}, \mathbf{u}_{2,i-1}) = 1, 2 \leq i \leq n-2\}|;$

$q \leftarrow |\{i \mid \sigma(\mathbf{u}_{3,i}, \mathbf{u}_{3,i-1}) = 1, 2 \leq i \leq n-2\}|;$

for $j = 4$ **to** n **do**

$\gamma_j \leftarrow |\{i \mid \sigma(\mathbf{u}_{j,i}, \mathbf{u}_{j,i-1}) = 1, \text{ for}$
 $2 \leq i \leq n-2\}|;$

if $\gamma_j = \min_{4 \leq j' \leq j} \{\gamma_{j'}\}$

then $\mathbf{u}' \leftarrow \mathbf{u}_{j,1};$

$r \leftarrow \gamma_j;$

```

if  $r = 0$ 
  then if  $p = 0$  and  $q = 0$ 
    then  $T \leftarrow \{\mathbf{u}_{2,1}, \mathbf{u}_{3,1}, \mathbf{u}'\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else if  $p = 0$  and  $q \neq 0$ 
    then  $T \leftarrow \{\mathbf{u}_{2,1}, \mathbf{u}'\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else if  $p \neq 0$  and  $q = 0$ 
    then  $T \leftarrow \{\mathbf{u}_{3,1}, \mathbf{u}'\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else
     $T \leftarrow \{\mathbf{u}'\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
else
  if  $p = 0$  and  $q = 0$ 
    then  $T \leftarrow \{\mathbf{u}_{2,1}, \mathbf{u}_{3,1}\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else if  $p = 0$  and  $q \neq 0$ 
    then  $T \leftarrow \{\mathbf{u}_{2,1}\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else if  $p \neq 0$  and  $q = 0$ 
    then  $T \leftarrow \{\mathbf{u}_{3,1}\};$ 
    return  $\text{VOTE}(T, \mathbf{u});$ 
  else
    return 0;
End

```

With the definition and the property of the PMC diagnosis model, we have the following two lemmas.

Lemma 1. *Let $P = \langle p_1, p_2, p_3, \dots, p_r \rangle$ be a path. If p_1 is faulty, and $\sigma(p_i, p_{i-1}) = 0$ for every $2 \leq i \leq r$, then all the r vertices in P are faulty.*

Lemma 2. *Let $P = \langle p_1, p_2, p_3, \dots, p_r \rangle$ be a path. If $\sigma(p_i, p_{i-1}) = 1$ for some i in $2 \leq i \leq r$, then there exists at least one fault vertex in P .*

The following theorem ensures that a vertex in a star graph can be diagnosed correctly with the algorithm **SDA** under the PMC model.

Theorem 1. *Suppose that $n \geq 4$. If F is a faulty vertex set in S_n with $|F| \leq n-1$, then the fault/fault-free status of vertex $\mathbf{u}$ can be identified correctly with the algorithm $\text{SDA}(S_n, \mathbf{u})$.*

Proof. For $2 \leq i \leq n$, let $\sigma_i = (\sigma(\mathbf{u}_{i,2}, \mathbf{u}_{i,1}), \sigma(\mathbf{u}_{i,3}, \mathbf{u}_{i,2}), \dots, \sigma(\mathbf{u}_{i,n-2}, \mathbf{u}_{i,n-3}))$ and $e_0 = (z_1, z_2, \dots, z_{n-3})$ where $z_{i'} = 0$ for every $1 \leq i' \leq n-3$. We consider the following cases.

Case 1: Suppose that there exists some k such that $\sigma_k = e_0$ for $3 \leq k \leq n-1$.

Subcase 1.1: Suppose that $\sigma_2 = e_0$ and $\sigma_3 = e_0$. We claim that $\mathbf{u} \notin F$ if at least two of $\sigma(\mathbf{u}_{2,1}, \mathbf{u})$, $\sigma(\mathbf{u}_{3,1}, \mathbf{u})$ and $\sigma(\mathbf{u}_{k,1}, \mathbf{u})$ are zero. Without loss of generality, suppose that $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 0$ and $\sigma(\mathbf{u}_{3,1}, \mathbf{u}) = 0$. We assume that $\mathbf{u} \in F$ by contradiction. Thus $\{\mathbf{u}_{2,1}, \mathbf{u}_{3,1}\} \subset F$. By Lemma 1, $|F| \geq 2(n-2) + 1 > n-1$ if $n \geq 4$, which contradicts the assumption that $|F| \leq n-1$. Now we assume that $\mathbf{u} \notin F$, and at least two of $\sigma(\mathbf{u}_{2,1}, \mathbf{u})$, $\sigma(\mathbf{u}_{3,1}, \mathbf{u})$ and $\sigma(\mathbf{u}_{k,1}, \mathbf{u})$ are one. Without loss of generality, suppose that $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 1$ and $\sigma(\mathbf{u}_{3,1}, \mathbf{u}) = 1$. Thus $\{\mathbf{u}_{2,1}, \mathbf{u}_{3,1}\} \subset F$. By Lemma 1, $|F| \geq 2(n-2) > n-1$ if $n \geq 4$, which contradicts the assumption that $|F| \leq n-1$.

Subcase 1.2: Suppose that $\sigma_2 = e_0$ and $\sigma_3 \neq e_0$. We claim that $\mathbf{u} \notin F$ if at least one of $\sigma(\mathbf{u}_{2,1}, \mathbf{u})$ and $\sigma(\mathbf{u}_{k,1}, \mathbf{u})$ is zero. Without loss of generality, suppose that $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 0$. We assume that $\mathbf{u} \in F$ by contradiction. Thus $\mathbf{u}_{2,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 1 + 1 > n-1$, which contradicts the assumption that $|F| \leq n-1$. Now we assume that $\mathbf{u} \notin F$, $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 1$ and $\sigma(\mathbf{u}_{k,1}, \mathbf{u}) = 1$. Thus $\{\mathbf{u}_{2,1}, \mathbf{u}_{k,1}\} \subset F$. By Lemma 1 and Lemma 2, $|F| \geq 2(n-2) + 1 > n-1$ if $n \geq 4$, which contradicts the assumption that $|F| \leq n-1$.

Subcase 1.3: Suppose that $\sigma_2 \neq e_0$ and $\sigma_3 = e_0$. This case is similar as Subcase 1.2 by interchanging the roles of $\mathbf{u}_{2,1}$ and $\mathbf{u}_{3,1}$.

Subcase 1.4: Suppose that $\sigma_2 \neq e_0$ and $\sigma_3 \neq e_0$. We claim that $\mathbf{u} \notin F$ if $\sigma(\mathbf{u}_{k,1}, \mathbf{u}) = 0$. We assume that $\mathbf{u} \in F$ by contradiction. Thus $\mathbf{u}_{k,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 2 + 1 > n-1$, which contradicts the assumption that $|F| \leq n-1$. Now we assume that $\mathbf{u} \notin F$ and $\sigma(\mathbf{u}_{k,1}, \mathbf{u}) = 1$. Thus $\mathbf{u}_{k,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 2 > n-1$, which contradicts the assumption that $|F| \leq n-1$.

Case 2: Suppose that $\sigma_k \neq e_0$ for $3 \leq k \leq n-1$.

Subcase 2.1: Suppose that $\sigma_2 = e_0$ and $\sigma_3 = e_0$. We claim that $\mathbf{u} \notin F$ if at least one of $\sigma(\mathbf{u}_{2,1}, \mathbf{u})$ and $\sigma(\mathbf{u}_{3,1}, \mathbf{u})$ is zero. Without loss of generality, suppose that $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 0$. We assume that $\mathbf{u} \in F$ by contradiction. Thus $\mathbf{u}_{2,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 1 + 1 > n-1$, which contradicts the assumption that $|F| \leq n-1$. Now we assume that $\mathbf{u} \notin F$, $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 1$ and $\sigma(\mathbf{u}_{3,1}, \mathbf{u}) = 1$. Thus $\{\mathbf{u}_{2,1}, \mathbf{u}_{3,1}\} \subset F$. By Lemma 1 and

Lemma 2, $|F| \geq 2(n-2) + 1 > n-1$ if $n \geq 4$, which contradicts the assumption that $|F| \leq n-1$.

Subcase 2.2: Suppose that $\sigma_2 = e_0$ and $\sigma_3 \neq e_0$. We claim that $\mathbf{u} \notin F$ if $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 0$. We assume that $\mathbf{u} \in F$ by contradiction. Thus $\mathbf{u}_{2,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 2 + 1 > n-1$, which contradicts the assumption that $|F| \leq n-1$. Now we assume that $\mathbf{u} \notin F$ and $\sigma(\mathbf{u}_{2,1}, \mathbf{u}) = 1$. Thus $\mathbf{u}_{2,1} \in F$. By Lemma 1 and Lemma 2, $|F| \geq (n-2) + 2 > n-1$, which contradicts the assumption that $|F| \leq n-1$.

Subcase 2.3: Suppose that $\sigma_2 \neq e_0$ and $\sigma_3 = e_0$. This case is similar as Subcase 2.2 by interchanging the roles of $\mathbf{u}_{2,1}$ and $\mathbf{u}_{3,1}$.

Subcase 2.4: Suppose that $\sigma_2 \neq e_0$ and $\sigma_3 \neq e_0$. By Lemma 2, there exist $n-1$ fault vertices in $\{\mathbf{u}_{i,j} \mid 2 \leq i \leq n, 2 \leq j \leq n-2\}$. With the assumption that $|F| \leq n-1$, the vertex $\mathbf{u} \notin F$. $\square$

Now we describe the scheme that determining the fault status of a vertex $\mathbf{u}$ in a star graph S_n with three test rounds under the PMC model.

- The first test round: Perform the tests $(\mathbf{u}_{2,1}, \mathbf{u})$, $(\mathbf{u}_{2,2i+1}, \mathbf{u}_{2,2i})$ for every $1 \leq i \leq \lfloor \frac{n-1}{2} \rfloor - 1$, and $(\mathbf{u}_{j,2k}, \mathbf{u}_{j,2k-1})$ for every $3 \leq j \leq n$ and $1 \leq k \leq \lfloor \frac{n-1}{2} \rfloor$.
- The second test round: Perform the tests $(\mathbf{u}_{3,1}, \mathbf{u})$, $(\mathbf{u}_{2,2i}, \mathbf{u}_{2,2i-1})$ for every $1 \leq i \leq \lfloor \frac{n-1}{2} \rfloor$, and $(\mathbf{u}_{j,2k+1}, \mathbf{u}_{j,2k})$ for every $3 \leq j \leq n$ and $1 \leq k \leq \lfloor \frac{n-1}{2} \rfloor - 1$.
- The third test round: Let j be an integer in $4 \leq j \leq n$ such that the number $|\{i \mid \sigma(\mathbf{u}_{j,i}, \mathbf{u}_{j,i-1}) = 1, 2 \leq i \leq n-2\}|$ after the first two test rounds is minimal. Perform the test $(\mathbf{u}_{j,1}, \mathbf{u})$.

4 Conclusion

In this paper, we concern with the local diagnosability. We propose an adaptive local diagnosis algorithm **SDA** for a star graph with three test rounds. Future work will try to find some specific structure for the existing practical interconnection networks. Then design the efficient diagnosis algorithm of the system with the useful structure in accordance with various

conditions and diagnosis models.

References

- [1] S. B. Akers, B. Krishnameurthy, "A Group-Theoretic Model for Symmetric Interconnection Networks," *IEEE Trans. Comput.*, Vol. 38, pp. 555-566, 1989.
- [2] T. A. Bartic, J. Y. Mignolet, V. Nollet, T. Marescaux, D. Verkest, S. Vernalde, and R. Lauwereins, "Topology Adaptive Network-on-Chip Design and Implementation," *IEE Proc. Computers and Digital Techniques*, vol. 152, no. 4, July 2005.
- [3] A. Dahbura and G. Masson, "An $O(N^{2.5})$ Fault Identification Algorithm for Diagnosable Systems," *IEEE Trans. Computers*, Vol. 33, no. 6, pp. 486-492, 1984.
- [4] A. D. Friedman and L. Simoncini, "System-Level Fault Diagnosis," *The Computer J.*, Vol. 13, no. 3, pp. 47-53, 1980.
- [5] S. Y. Hsieh and Y.-S. Chen, "Strongly Diagnosable Product Networks under the Comparison Diagnosis Model," *IEEE Trans. on Computers*, Vol. 57, no. 6, pp. 721-732, 2008.
- [6] G. H. Hsu and J. J. M. Tan, "A Local Diagnosability Measure for Multiprocessor Systems," *IEEE Trans. Parallel and Distributed Systems*, Vol. 18, no. 5, pp. 598-607, 2007.
- [7] L. H. Hsu, C. K. Lin, *Graph Theory and Interconnection Networks*, CRC Press, 2008.
- [8] C. K. Lin, S. L. Peng, J. J. M. Tan, T. L. Kung, and L. H. Hsu, "The Diagnosability of g-Good-Neighbor Conditional-Faulty Hypercube Under PMC Model," *Proc. the 2010 International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA'10)*, vol. II, pp. 494-499, July 2010.
- [9] P. P. Pande, C. Grecu, M. Jones, A. Ikonov, and R. Saleh, "Performance Evaluation and Design Trade-Offs for Network-on-Chip Interconnect Architectures," *IEEE Trans. Computers*, Vol. 54, no. 8, pp. 1025-1040, 2005.
- [10] F. P. Preparata, G. Metze, and R. T. Chien, "On the Connection Assignment Problem of

- Diagnosis Systems,” *IEEE Trans. Electronic Computers*, Vol. 16, no. 12, pp. 848-854, 1967.
- [11] H. H. Wang and D. R. Duh, “Design of a Pessimistic Diagnosis Algorithm for t -ary Hypercube Multicomputer Systems,” Proc. 2009 Conf. on Information Technology and Applications in Outlying Islands, Jinning Township, Kinmen, Taiwan, pp. 285-291, May 22-24, 2009.